

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1_AT3 — VISUALIZATION DESIGN EXERCISE

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO1 – Precision (BL2)	Assessment:	Assessment Tool 3 – Visualization Design
Weightage:	40% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO1 Statement:	<i>Understand the fundamental concepts, principles, and techniques of data visualization.(BL2)</i>		
Student Name:	RAGUL.M	Reg. No.:	192524066
Section / Batch:	SLOT C	Date:	30/07/2026

Code of Conduct

I RAGUL.M/192524066 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: RAGUL.M

Instructions

- This assessment contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Annexure A – VISUALIZATION DESIGN

ASSESSMENT RUBRIC

AT3 — Visualization Design Exercise

CO1 · BT3: Apply ·

This rubric is used to assess student responses to the AT3 Visualization Design Exercise problems, in which students design and justify a complete visualization (chart type, aesthetic mappings, scales, coordinate systems, and/or color scheme) for a given dataset or design brief.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of chart types, aesthetic mappings, scales, coordinate systems, and color scales, and correctly identifies which apply to the given design scenario.	Good understanding of the relevant design concepts, with only minor gaps in selecting appropriate chart types, aesthetic mappings, or scales.	Basic understanding of design concepts, but with some misconceptions about appropriate chart types, aesthetics, or scales for the scenario.	Poor understanding of core design concepts; proposes chart types or mappings that are fundamentally unsuited to the data.
Explanation	25%	Provides detailed, well-reasoned justification for every design decision, explicitly connecting each choice to the properties of the data and the scenario.	Provides clear justification for design decisions, with adequate reasoning for most sub-questions.	Justification is limited or superficial; design choices are stated without sufficient reasoning.	Little or no justification provided; design choices appear arbitrary or unexplained.
Application	20%	Effectively applies visualization design principles to produce a complete, appropriate, and well-reasoned design solution for the specific scenario.	Applies design principles to produce an appropriate solution, with only minor errors or missed refinements.	Limited application of design principles; the proposed design partially addresses the scenario but has notable weaknesses.	Fails to apply appropriate design principles; the proposed design does not suit the scenario or data.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Organization	15%	The design proposal is clearly structured, addressing each sub-question (a, b, c) in a logical sequence with coherent flow throughout.	The proposal is organized and addresses each sub-question, with only minor issues in flow or clarity.	Basic structure; sub-questions are addressed but the proposal lacks clarity or a logical sequence.	Poorly organized; the proposal is incoherent, and sub-questions are left unaddressed or answered out of order.
Accuracy	10%	All design choices — chart type, aesthetic mapping, scale, coordinate system, and/or color scheme — are completely accurate and well-suited to the data.	Mostly accurate design choices, with only minor omissions or a small error in one part of the proposal.	Partially correct; some design elements are missing, mismatched to the data, or incorrect.	Design choices are largely incorrect or inappropriate, with major gaps in matching the visualization to the data.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

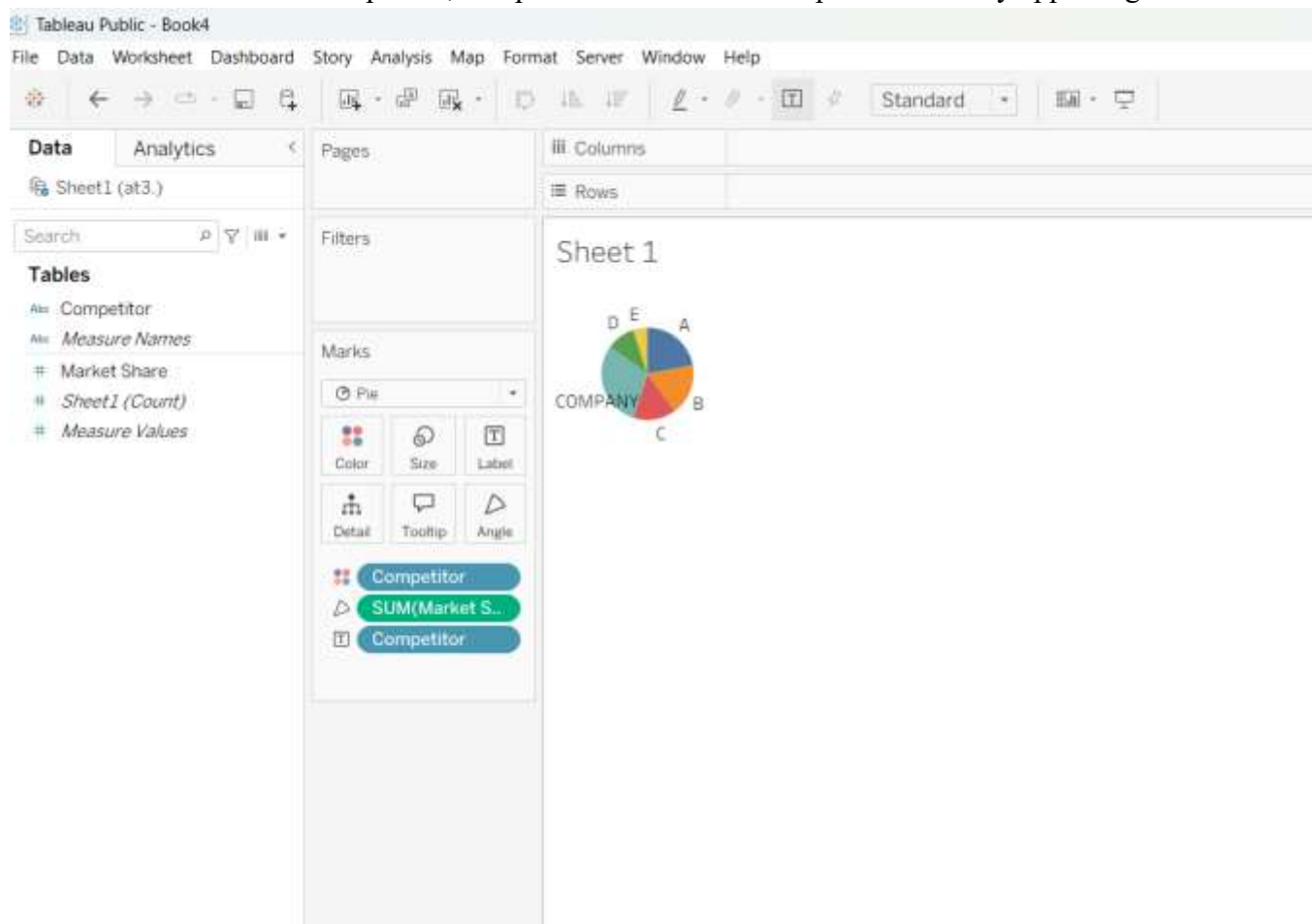
Problem 9 – Market Share Analysis (Pie Chart)

Introduction

Market share is one of the most important indicators used to measure a company's position in a competitive market. It represents the percentage of total sales captured by a company compared with its competitors. Visualizing market share helps businesses understand their performance, identify market leaders, and analyze competition. A pie chart is an ideal visualization because it clearly shows how the total market is divided among different companies.

Objective

The objective of this visualization is to analyze and compare the market share of a company with five competitors. The chart helps identify the company with the highest and lowest market share, understand the contribution of each competitor, and present the data in a simple and visually appealing format.



Observation

The pie chart divides the entire market into individual slices, where each slice represents a competitor's market share. Companies with higher market share are displayed as larger slices, while those with lower

market share appear as smaller slices. Different colors help distinguish each competitor clearly. The visualization makes it easy to understand the overall market distribution and compare the contribution of each company at a glance.

Description

The dataset containing competitor names and market share percentages was imported into Tableau. The Competitor field was placed on Color and Label, while Market Share (%) was assigned to Angle. The Marks card was changed to Pie, and labels were enabled to display the percentage values. Appropriate colors and titles were added to improve readability and presentation.

Conclusion

The pie chart effectively represents market share because it emphasizes the relationship between individual competitors and the total market. It allows users to quickly identify market leaders, compare competitors, and understand the overall market structure. This visualization supports business decision-making and competitive analysis.

Problem 19 – Patient Wait Time Heatmap

Introduction

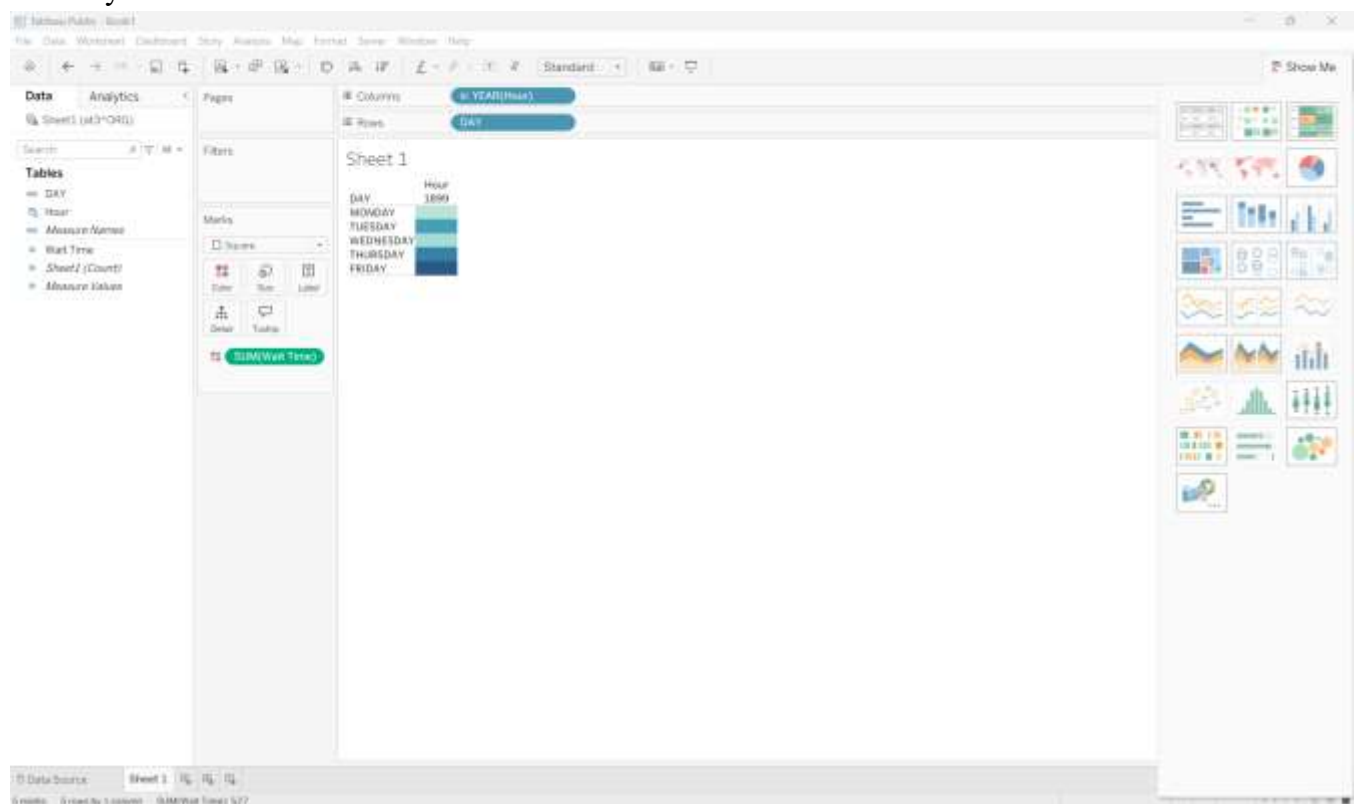
Patient waiting time is an important measure of healthcare service quality. Long waiting times may reduce patient satisfaction and indicate inefficient resource allocation. A heatmap is an effective visualization technique because it uses color intensity to represent different waiting times, making it easy to identify busy and less busy periods across days and hours.

Objective

The objective of this visualization is to analyze patient waiting times recorded at different hours throughout the week. It helps identify peak waiting periods, compare waiting times across different days, and support better planning of hospital staff and resources to improve patient care.

Observation

The heatmap displays waiting times using different color intensities. Darker shades represent longer waiting times, while lighter shades indicate shorter waiting times. By observing the color patterns, users can quickly identify the busiest hours and days. The visualization also highlights trends and variations that may not be obvious in a table of numerical values.



Description

The dataset containing Day, Hour, and Wait Time (Minutes) was imported into Tableau. The Day field was placed on the Rows shelf, the Hour field on the Columns shelf, and Wait Time (Minutes) was assigned to Color. The Marks card was set to Square to generate a heatmap. A sequential color palette was applied, where lighter colors represent lower waiting times and darker colors represent higher waiting times. Titles, labels, and legends were included to improve readability and interpretation.

Conclusion

The heatmap provides a clear and effective visualization of patient waiting times across different days and hours. It helps hospital administrators identify peak waiting periods, improve staff scheduling, optimize resource allocation, and enhance the overall patient experience. The visualization supports informed decision-making and contributes to better healthcare management.